



ADITYA UNIVERSITY

Data Analysis Essentials Cornerstone Project

First Review

Student Performance Data Analysis System

Presented by

D. Harsha Kumar (25B11AI257)

K.S.R. Harsha Vardhan (25B11AI531)

P.S. Sai Pavan (25B11AI975)

U.P.C. Manikanta (25B11AIC20)

Department of Artificial Intelligence & Machine Learning

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The **Student Performance Data Analysis System** is a Python-based data analytics project that helps analyze student academic records and understand the factors affecting academic performance. The project uses the **Pandas** library to load and process student datasets, performs data cleaning to handle missing or inconsistent values, calculates average marks and grades, analyzes attendance records, and visualizes performance trends using **Matplotlib** charts. It demonstrates a complete data analysis workflow, enabling users to gain meaningful insights into student performance and providing a strong foundation for future Data Science and Machine Learning applications.

Educational institutions manage the academic records of many students.

Student data includes:

- Student Name
- Roll Number
- Attendance
- Subject Marks
- Average Marks
- Grade
- Subject-wise Performance

Manual analysis is:

- Time consuming
 - Error prone
 - Difficult for large datasets
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- Using **Python Data Analysis**, we can automatically clean, analyze and visualize the data to assist Students Data.

Existing System:

Student performance records are generally maintained manually.

Data is usually stored in:

Excel Sheets

CSV Files

Paper Records

Problems include:

Slow processing

Human errors

Duplicate records

Difficult to identify performance trends

No automatic visualization

Difficult to compare student performance

Limitations of Existing System:

- Large amount of manual work
- Missing data affects decision making
- Difficult to compare applicants
- No graphical insights Poor prediction accuracy

Proposed System :

Develop a **Student Performance Data Analysis System** using Python.

The system performs:

- Load dataset using Pandas
- Explore the dataset
- Clean missing or duplicate data
- Calculate average marks
- Assign grades automatically
- Generate Graphs and visualizations
- Analyze attendance records
- Identify top-performing and weak-performing students
- Generate performance reports

Advantages

- Faster data processing
- Better data quality
- Automatic cleaning
- Easy visualization
- Better understanding of customer profiles

Feature	Description
Student_ID	Unique Student Number
Name	Student Name
Gender	Male/Female
Age	Student Age
Maths	Number of dependents
Physics	Physics Marks
Chemistry	Chemistry Marks
English	English
Computer_Science	Computer Science Marks
Attendance	Attendance Percentage
Average	Average Marks
Grade	Student Grade

Hardware Requirements

- Processor: Intel i3 or above
- RAM: Minimum 4 GB
- Storage: 20 GB
- Input Devices: Keyboard, Mouse
- Output Device: Monitor

Software Requirements

- Operating System: Windows / Linux
- Programming Language: python
- IDE: Google Collab/ Jupyter Notebook
- Libraries: Pandas, Numpy, Matplotlib
- GitHub (for version control)

Modules:

Module 1

- Dataset Loading
- Import libraries
- Read CSV file
- Load student dataset
- Display first rows

Module 2

- Data Exploration
- Dataset shape
- Column names
- Data types
- Missing values
- Statistical summary

Module 3

- Data Cleaning
- Fill missing values
- Remove duplicates records
- Handle incorrect data

Module 4

Data Preprocessing

- Calculate average marks
- Assign grades
- Sort student records
- Filter student performance

Module 5

Exploratory Data Analysis

Visualize

- Subject-wise average marks
- Attendance Analysis
- Grade distribution

Module 6

Performance Analysis

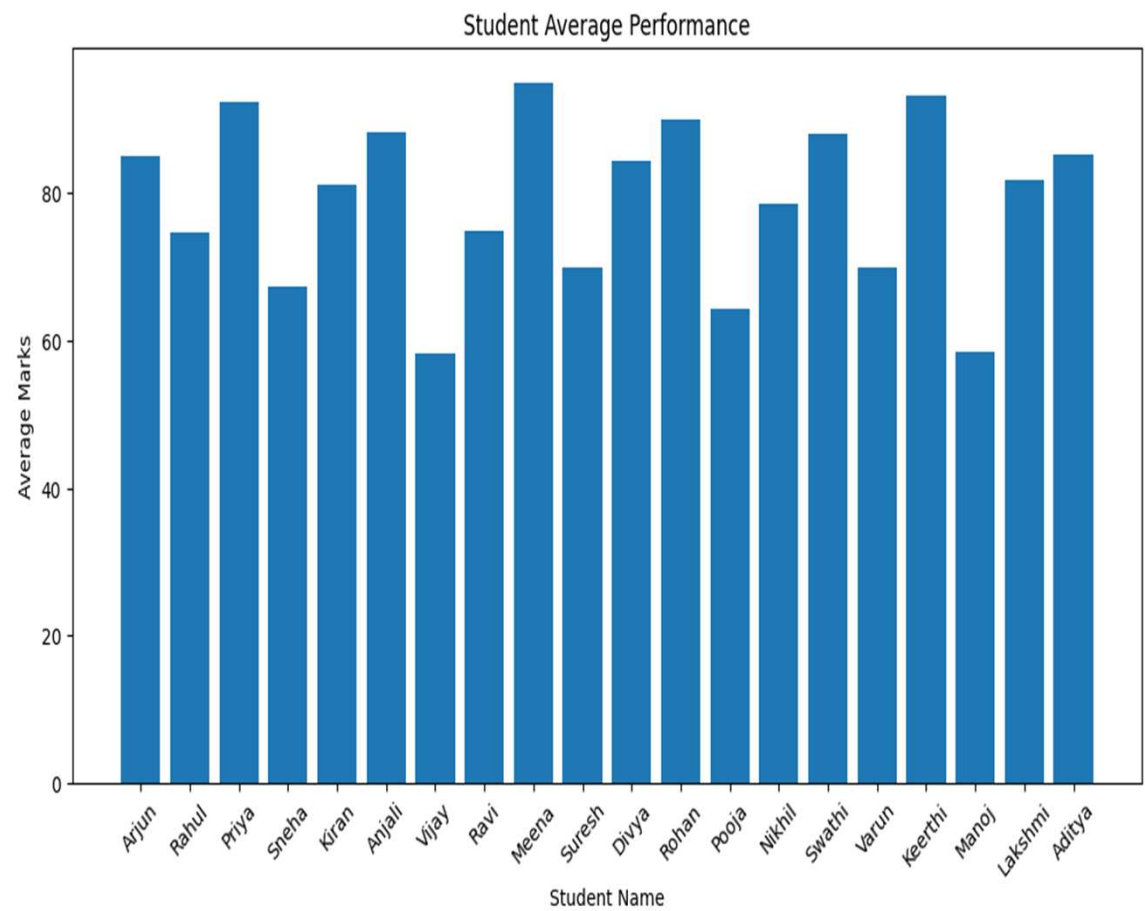
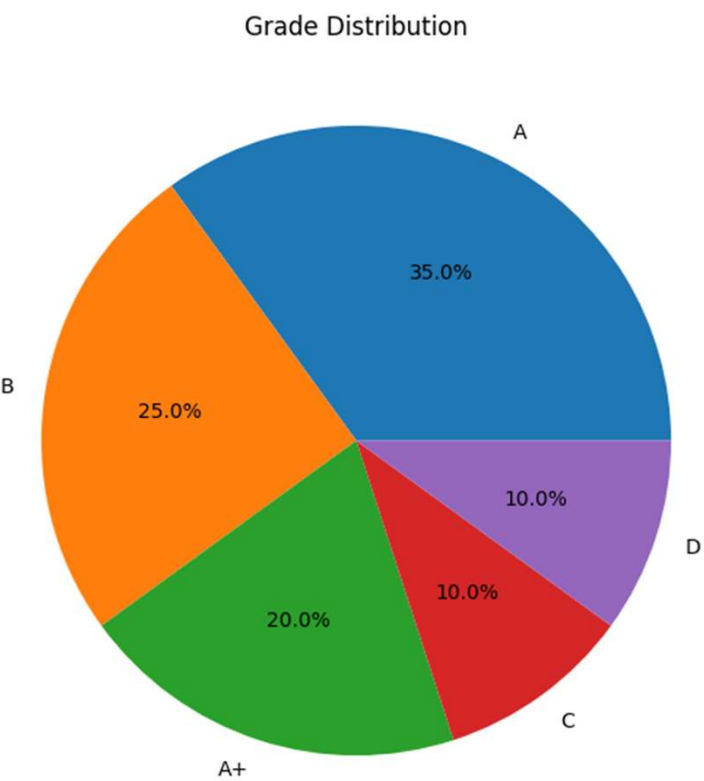
- Top-performing students
- Weak-performing students
- Best-performing student
- Attendance impact on performance

The system displays:

- Dataset Information
- Missing Value Report
- Cleaned Dataset
- Student Performance Report
- Average Marks
- Grade Assignment
- Attendance Analysis
- Charts and Graphs
- Subject-wise Performance
- Topper Information
- Performance Insights

Expected Outputs

Visual Data:



CONCLUSION

The **Student Performance Data Analysis System** demonstrates the complete data analysis pipeline using Python. The project helps students understand:

- The project helps students understand
 - Data Collection
 - Data Cleaning
 - Data Analysis
 - Data Visualization
 - Performance Evaluation
 - Report Generation
- It provides valuable insights into student academic performance and creates a strong foundation for future **Data Science, Machine Learning, and Educational Analytics** projects.

THANK YOU